

OKI semiconductor

MSM3732 AS/RS

32,768-BIT DYNAMIC RANDOM ACCESS MEMORY

GENERAL DESCRIPTION

The Oki MSM3732H/L is a fully decoded, dynamic NMOS random access memory organized as 32,768 one-bit words. The design is optimized for high-speed, high performance applications such as mainframe memory, buffer memory, peripheral storage and environments where low power dissipation and compact layout is required.

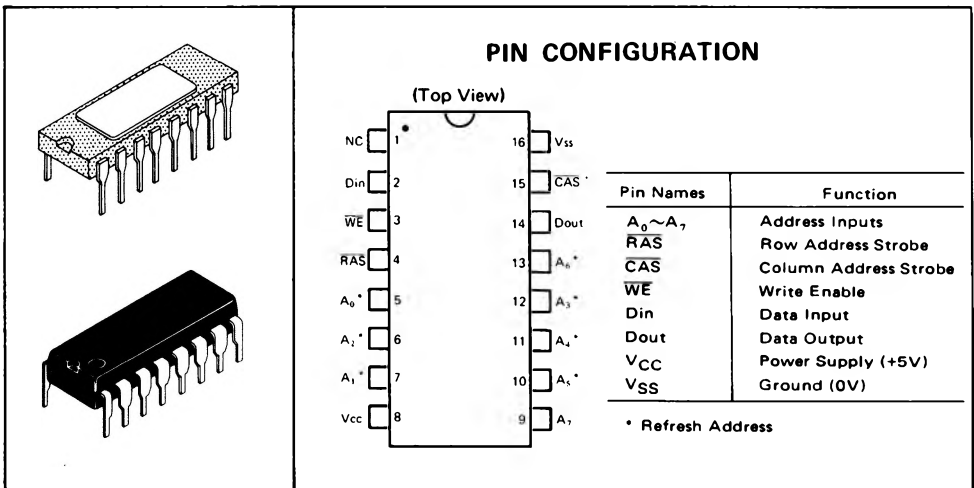
Multiplexed row and column address inputs permit the MSM3732 to be housed in a standard 16 pin DIP.

The MSM3732 is fabricated using silicon gate NMOS and Oki's advanced Double-Layer Polysilicon process. This process, coupled with single-transistor memory storage cells, permits maximum circuit density and minimum chip size. Dynamic circuitry is employed in the design, including the sense amplifiers.

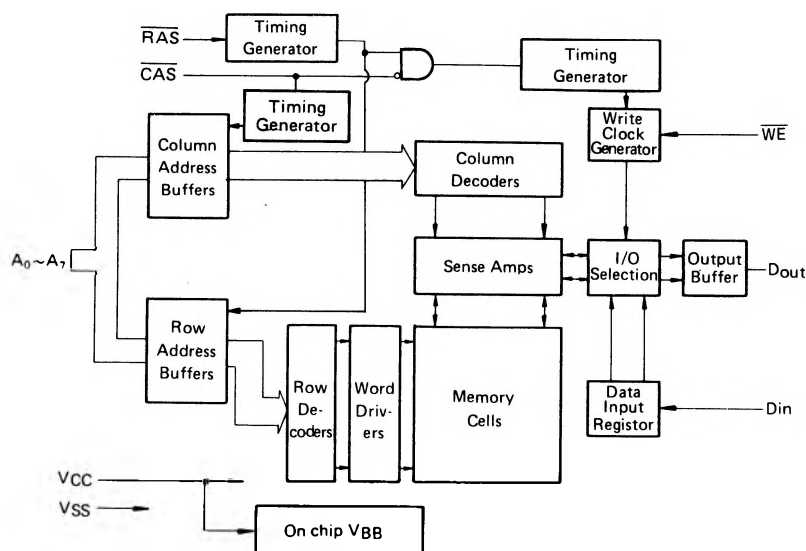
Clock timing requirements are noncritical, and power supply tolerance is very wide. All inputs and output are TTL compatible.

FEATURES

- 32,768 x 1 RAM, 16 pin package
- Silicon-gate, Double Poly NMOS, single transistor cell
- Row access time,
150 ns max (MSM3732H/L-15)
200 ns max (MSM3732H/L-20)
- Cycle time,
270 ns min (MSM3732H/L-15)
330 ns min (MSM3732H/L-20)
- Low power: 248 mW active,
28 mW max standby
- Single +5V Supply, $\pm 10\%$ tolerance
- All inputs TTL compatible, low capacitive load
- Three-state TTL compatible output
- "Gated" CAS
- 128 refresh cycles/2 ms
- Common I/O capability using "Early Write" operation
- Output unlatched at cycle end allows extended page boundary and two-dimensional chip select
- Read-Modify-Write, RAS-only refresh, and Page-Mode capability
- On-chip latches for Addresses and Data-in
- On-chip substrate bias generator for high performance



FUNCTIONAL BLOCK DIAGRAM



ABSOLUTE MAXIMUM RATINGS (See Note)

Rating	Symbol	Value	Unit
Voltage on any pin relative to V _{SS}	V _{IN} , V _{OUT}	-1 to +7	V
Voltage on V _{CC} supply relative to V _{SS}	V _{CC}	-1 to +7	V
Operating temperature	T _{opr}	0 to 70	°C
Storage temperature	T _{stg}	-55 to +150	°C
Power dissipation	P _D	1.0	W
Short circuit output current		50	mA

Note: Permanent device damage may occur if ABSOLUTE MAXIMUM RATINGS are exceeded. Functional operation should be restricted to the conditions as detailed in the operational sections of this data sheet. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

RECOMMENDED OPERATING CONDITIONS

(Referenced to V_{SS})

Parameter	Symbol	Min.	Typ.	Max.	Unit	Operating Temperature
Supply Voltage	V _{CC}	4.5	5.0	5.5	V	0°C to +70°C
	V _{SS}	0	0	0	V	
Input High Voltage, all inputs	V _{IH}	2.4		6.5	V	
Input Low Voltage, all inputs	V _{IL}	-1.0		0.8	V	

DC CHARACTERISTICS

(Recommended operating conditions unless otherwise noted.)

Parameter	Symbol	Min.	Max.	Unit	Notes
Operating Current* Average power supply current (RAS, CAS cycling; t _{RC} = min.)	I _{CC1}		45	mA	
Standby Current Power supply current (RAS = CAS = V _{IH})	I _{CC2}		5.0	mA	
Refresh Current* Average power supply current (RAS cycling, CAS = V _{IH} ; t _{RC} = min.)	I _{CC3}		35	mA	
Page Mode Current* Average power supply current (RAS = V _{IL} , CAS cycling; t _{PC} = min.)	I _{CC4}		42	mA	
Input Leakage Current Input leakage current, any input (0V ≤ V _{IN} ≤ 5.5V, all other pins not under test = 0V)	I _{LI}	-10	10	μA	
Output Leakage Current (Data out is disabled, 0V ≤ V _{OUT} ≤ 5.5V)	I _{LO}	-10	10	μA	
Output Levels Output high voltage (I _{OH} = -5 mA) Output low voltage (I _{OL} = 4.2 mA)	V _{OH} V _{OL}	2.4	0.4	V V	

Note*: ICC is dependent on output loading and cycle rates. Specified values are obtained with the output open.

CAPACITANCE

(T_a = 25°C, f = 1 MHz)

Parameter	Symbol	Typ.	Max.	Unit
Input Capacitance (A ₀ ~ A ₇ , D _{IN})	C _{IN1}	4.5	5	pF
Input Capacitance (RAS, CAS, WE)	C _{IN2}	7	10	pF
Output Capacitance (D _{OUT})	C _{OUT}	5	7	pF

Capacitance measured with Boonton Meter.

AC CHARACTERISTICS

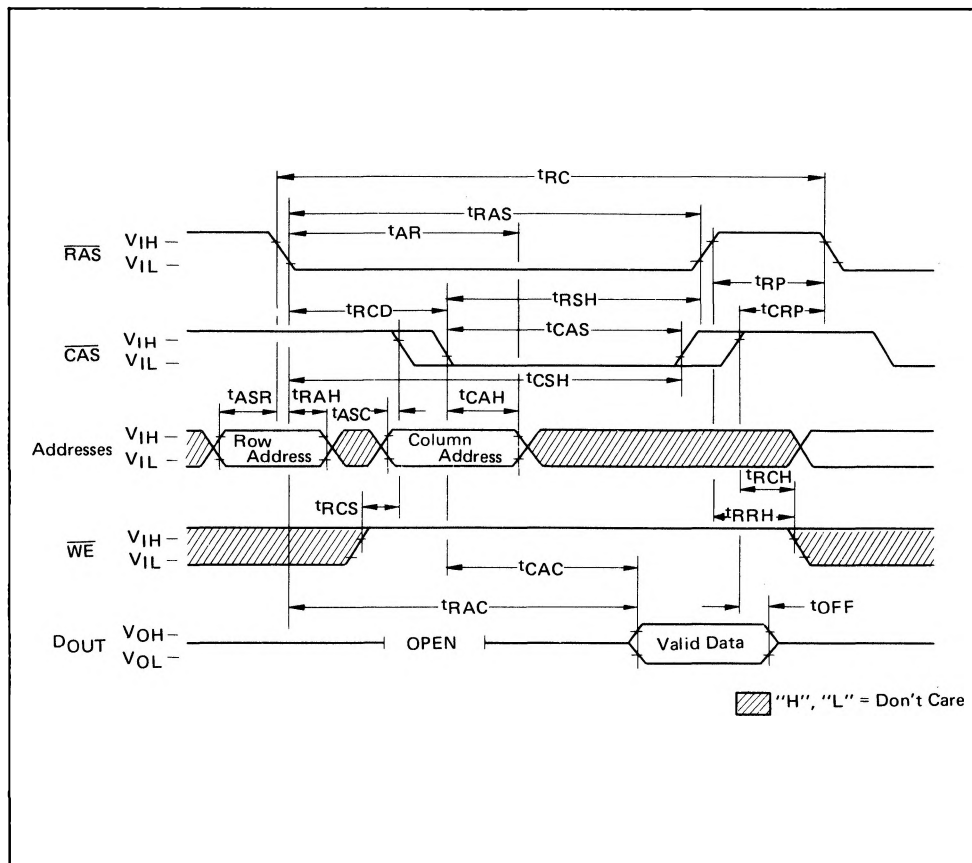
(Recommended operating conditions unless otherwise noted.)

Note 1,2,3

Parameter	Symbol	Units	MSM3732-15		MSM3732-20		Note
			Min.	Max.	Min.	Max.	
Refresh period	t _{REF}	ms		2		2	
Random read or write cycle time	t _{RC}	ns	270		330		
Read-write cycle time	t _{RWC}	ns	270		330		
Page mode cycle time	t _{PC}	ns	170		225		
Access time from $\overline{\text{RAS}}$	t _{RAC}	ns		150		200	4, 6
Access time from $\overline{\text{CAS}}$	t _{CAC}	ns		100		135	5, 6
Output buffer turn-off delay	t _{OFF}	ns	0	40	0	50	
Transition time	t _T	ns	3	35	3	50	
RAS precharge time	t _{RP}	ns	100		120		
RAS pulse width	t _{RAS}	ns	150	10,000	200	10,000	
RAS hold time	t _{RSH}	ns	100		135		
$\overline{\text{CAS}}$ precharge time	t _{CP}	ns	60		80		
$\overline{\text{CAS}}$ pulse width	t _{CAS}	ns	100	10,000	135	10,000	
$\overline{\text{CAS}}$ hold time	t _{CSH}	ns	150		200		
RAS to $\overline{\text{CAS}}$ delay time	t _{RCD}	ns	25	50	30	65	7
$\overline{\text{CAS}}$ to RAS precharge time	t _{CRP}	ns	0		0		
Row Address set-up time	t _{ASR}	ns	0		0		
Row Address hold time	t _{RAH}	ns	15		20		
Column Address set-up time	t _{ASC}	ns	0		0		
Column Address hold time	t _{CAH}	ns	45		55		
Column Address hold time referenced to RAS	t _{AR}	ns	95		120		
Read command set-up time	t _{RCS}	ns	0		0		
Read command hold time	t _{RCH}	ns	0		0		
Write command set-up time	t _{WCS}	ns	-10		-10		8
Write command hold time	t _{WCH}	ns	45		55		
Write command hold time referenced to RAS	t _{WCR}	ns	95		120		
Write command pulse width	t _{WP}	ns	45		55		
Write command to RAS lead time	t _{RWL}	ns	45		55		
Write command to $\overline{\text{CAS}}$ lead time	t _{CWL}	ns	45		55		
Data-in set-up time	t _{DS}	ns	0		0		
Data-in hold time	t _{DH}	ns	45		55		
Data-in hold time referenced to RAS	t _{DHR}	ns	95		120		
$\overline{\text{CAS}}$ to $\overline{\text{WE}}$ delay	t _{CWD}	ns	60		80		8
RAS to $\overline{\text{WE}}$ delay	t _{RWD}	ns	110		145		8
Read command hold time referenced to RAS	t _{RRH}	ns	20		25		

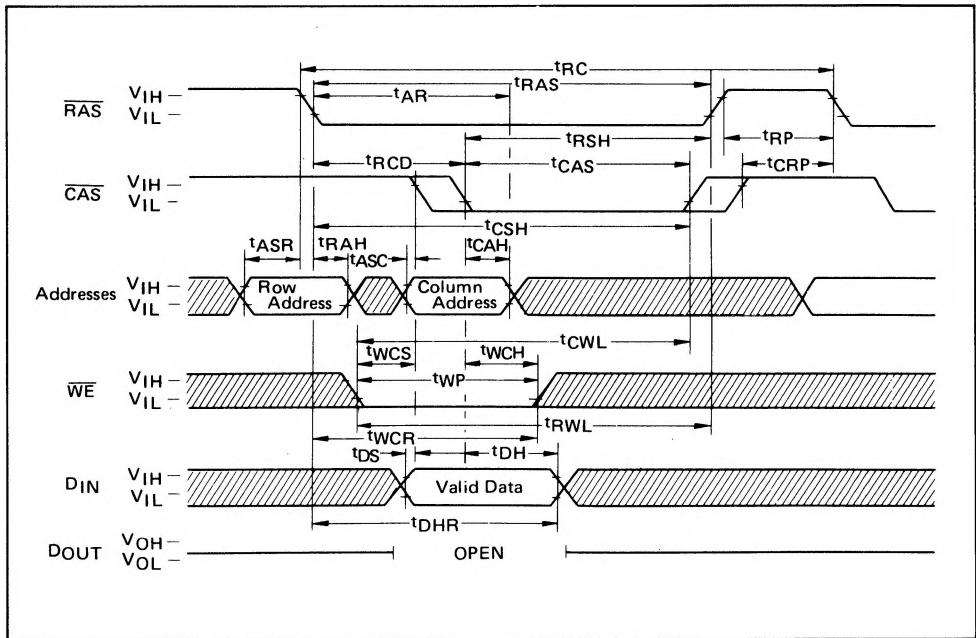
- NOTES:**
- 1) An initial pause of 100 μ s is required after power-up followed by any 8 $\overline{\text{RAS}}$ cycles (Examples; $\overline{\text{RAS}}$ only) before proper device operation is achieved.
 - 2) AC measurements assume $t_T = 5$ ns.
 - 3) V_{IH} (Min.) and V_{IL} (Max.) are reference levels for measuring timing of input signals. Also, transition times are measured between V_{IH} and V_{IL} .
 - 4) Assumes that $t_{RCD} < t_{RCD}(\text{max.})$.
If t_{RCD} is greater than the maximum recommended value shown in this table, t_{RAC} will increase by the amount that t_{RCD} exceeds the value shown.
 - 5) Assumes that $t_{RCD} < t_{RCD}(\text{max.})$
 - 6) Measured with a load circuit equivalent to 2 TTL loads and 100 pF.
 - 7) Operation within the $t_{RCD}(\text{max.})$ limit insures that $t_{RAC}(\text{max.})$ can be met. $t_{RCD}(\text{max.})$ is specified as a reference point only; if t_{RCD} is greater than the specified $t_{RCD}(\text{max.})$ limit, then access time is controlled exclusively by t_{CAC} .
 - 8) t_{WCS} , t_{CWD} and t_{RWD} are not restrictive operating parameters. They are included in the data sheet as electrical characteristics only; if $t_{WCS} \geq t_{WCS}(\text{min.})$, the cycle is an early write cycle and the data out pin will remain open circuit (high impedance) throughout the entire cycle; if $t_{CWD} \geq t_{CWD}(\text{min.})$ and $t_{RWD} > t_{RWD}(\text{min.})$ the cycle is read-write cycle and the data out will contain data read from the selected cell; if neither of the above sets of conditions is satisfied the condition of the data out (at access time) is indeterminate.

READ CYCLE TIMING

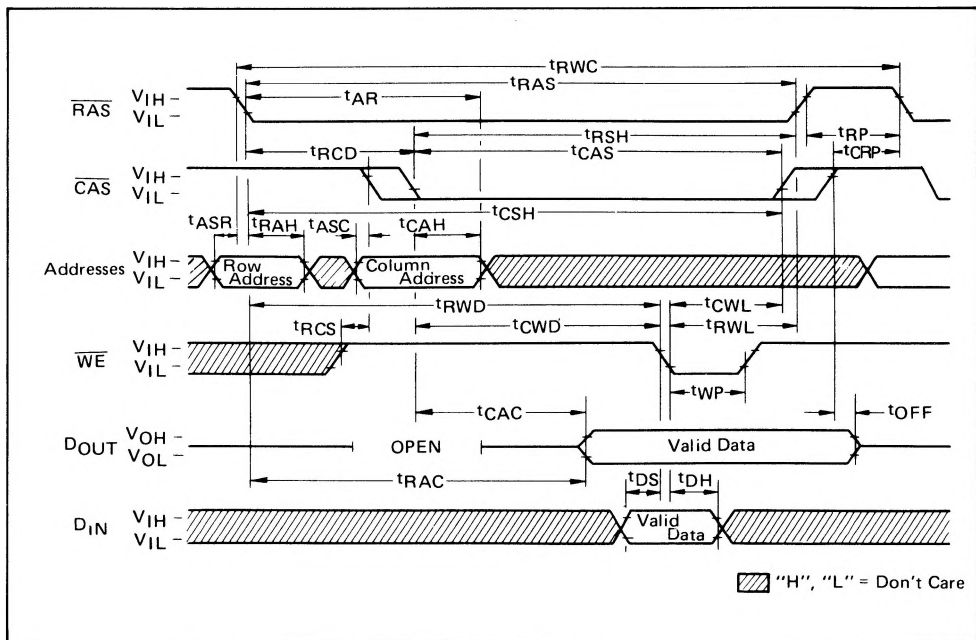


WRITE CYCLE TIMING

(EARLY WRITE)

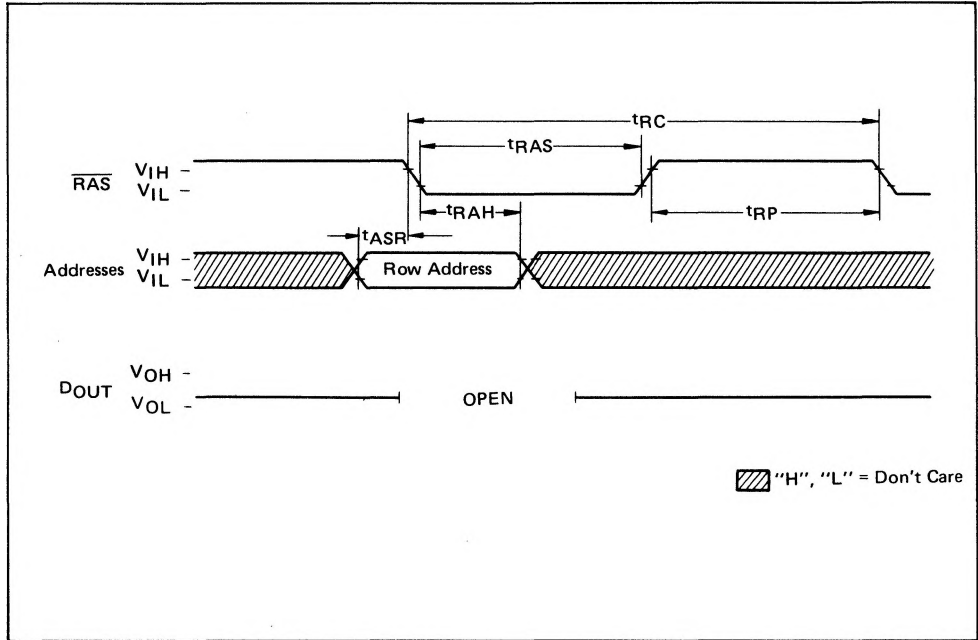


READ-WRITE/READ-MODIFY-WRITE CYCLE

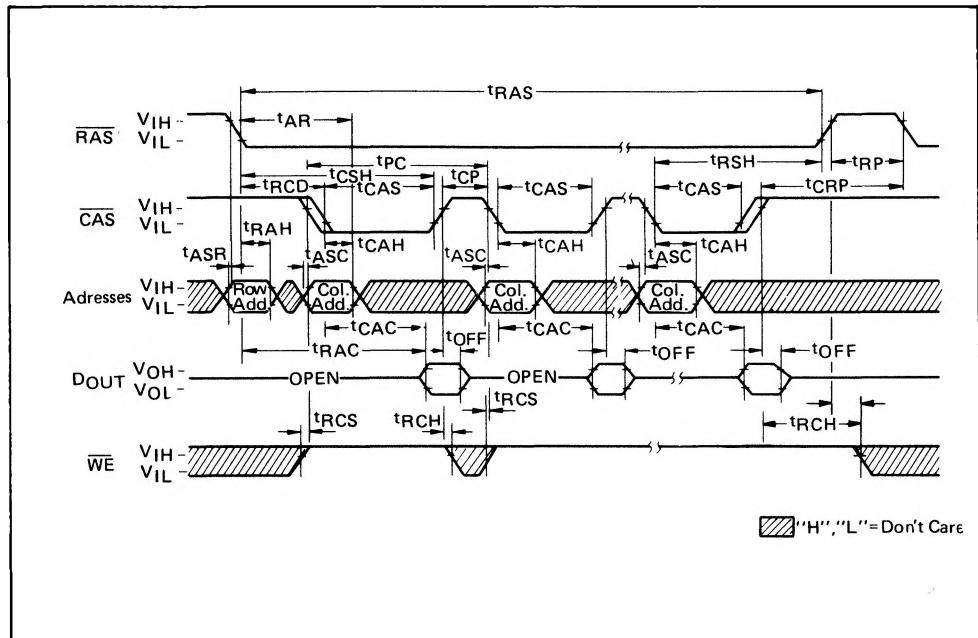


RAS ONLY REFRESH TIMING

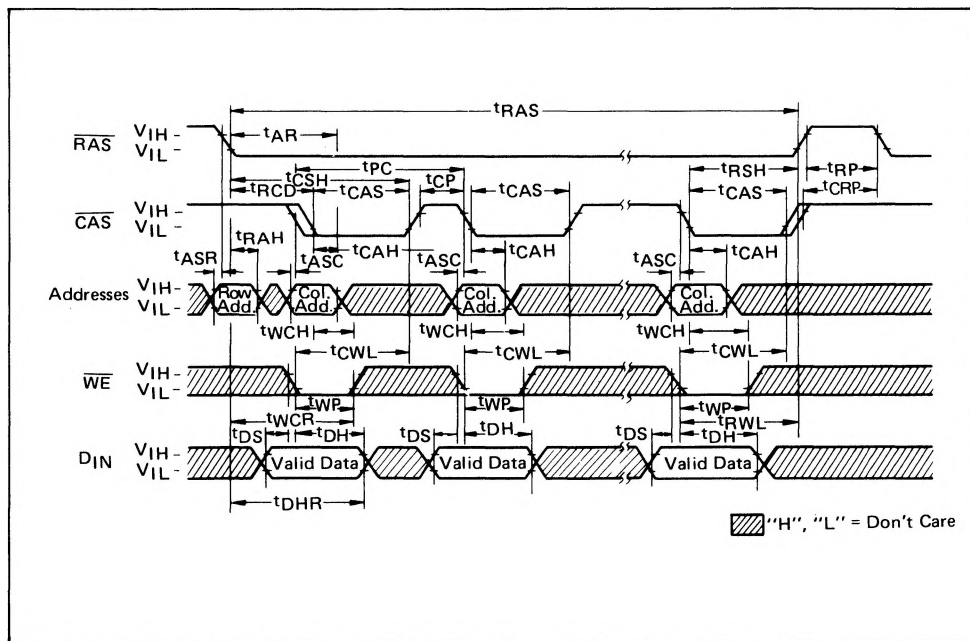
(CAS: V_{IH} , $\overline{WE}$ & DIN: Don't care)



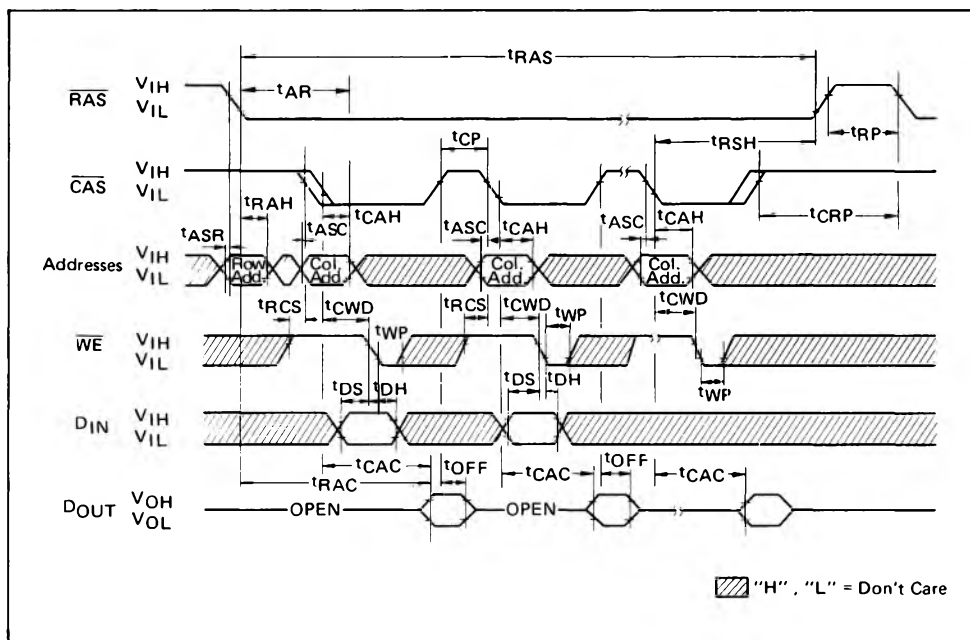
PAGE MODE READ CYCLE



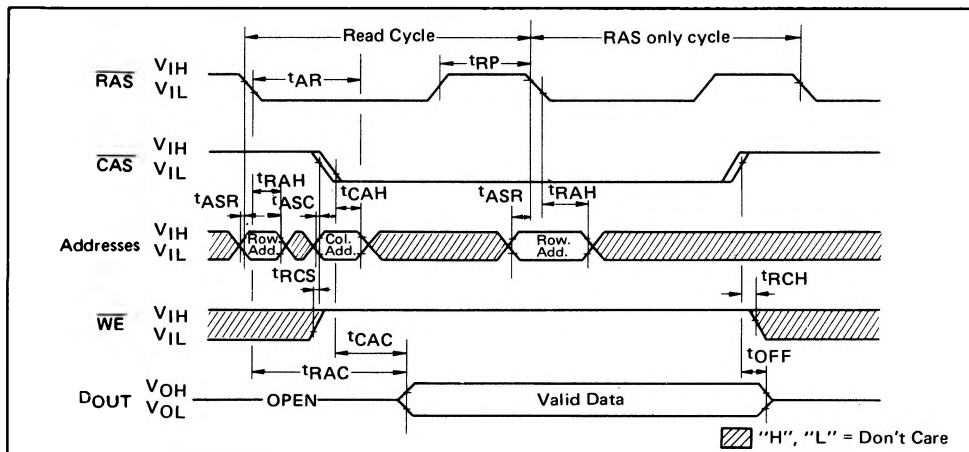
PAGE MODE WRITE CYCLE



PAGE MODE, READ-MODIFY-WRITE CYCLE



HIDDEN REFRESH



DESCRIPTION

Address Inputs:

A total of fifteen binary input address bits are required to decode any 1 of 32,768 storage cell locations within the MSM3732. Eight row-address bits are established on the input pins ($A_0 \sim A_7$) and latched with the Row Address Strobe ($\overline{RAS}$). The seven column-address bits (A_0 through A_6) are established on the input pins and latched with the Column Address Strobe ($\overline{CAS}$). All input addresses must be stable on or before the falling edge of $\overline{RAS}$. $\overline{CAS}$ is internally inhibited (or "gated") by $\overline{RAS}$ to permit triggering of $\overline{CAS}$ as soon as the Row Address Hold Time (t_{RAH}) specification has been satisfied and the address inputs have been changed from row-addresses to column-addresses.

One Column Address (A_7) has to be fixed at logic "0" (low level) for MSM3732L, and at logic "1" (high level) for MSM3732H.

Write Enable:

The read mode or write mode is selected with the $\overline{WE}$ input. A logic high (1) on $\overline{WE}$ dictates read mode; logic low (0) dictates write mode. Data input is disabled when read mode is selected.

Data Input:

Data is written into the MSM3732 during a write or read-write cycle. The last falling edge of $\overline{WE}$ or $\overline{CAS}$ is a strobe for the Data In (D_{IN}) register. In a write cycle, if $\overline{WE}$ is brought low (write mode) before $\overline{CAS}$, D_{IN} is strobed by $\overline{CAS}$, and the set-up and hold times are referenced to $\overline{CAS}$. In a read-write cycle, $\overline{WE}$ will be delayed until $\overline{CAS}$ has made its negative transition. Thus D_{IN} is strobed by $\overline{WE}$, and set-up and hold times are referenced to $\overline{WE}$.

Data Output:

The output buffer is three-state TTL compatible with

a fan-out of two standard TTL loads. Data-out is the same polarity as data-in. The output is in a high impedance state until $\overline{CAS}$ is brought low. In a read cycle, or read-write cycle, the output is valid after t_{RAC} from transition of $\overline{RAS}$ when t_{RCD} (max.) is satisfied, or after t_{CAC} from transition of $\overline{CAS}$ when the transition occurs after t_{RCD} (max.). Data remain valid until $\overline{CAS}$ is returned to a high level. In a write cycle the identical sequence occurs, but data is not valid.

Page Mode:

Page-mode operation permits strobing the row-address into the MSM3732 while maintaining $\overline{RAS}$ at a logic low (0) throughout all successive memory operations in which the row-address doesn't change. Thus the power dissipated by the negative going edge of $\overline{RAS}$ is saved. Further, access and cycle times are decreased because the time normally required to strobe a new row-address is eliminated.

Refresh:

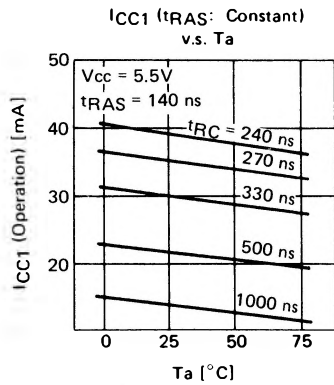
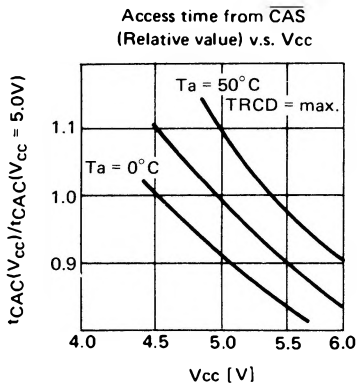
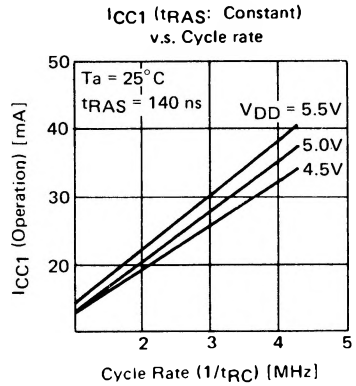
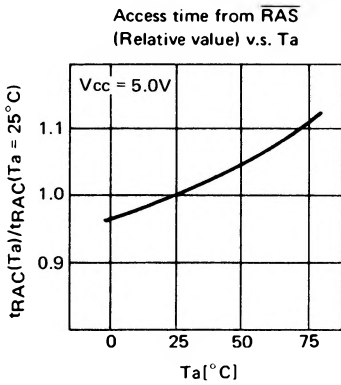
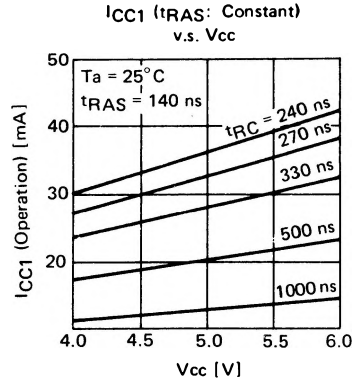
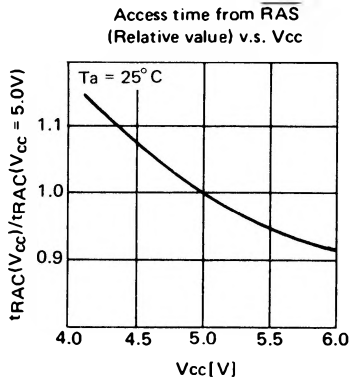
Refresh of the dynamic memory cells is accomplished by performing a memory cycle at each of the 128 row-addresses ($A_0 \sim A_6$) at least every two milliseconds. During refresh, either V_{IL} or V_{IH} is permitted for A_7 . $\overline{RAS}$ only refresh avoids any output during refresh because the output buffer is in the high impedance state unless $\overline{CAS}$ is brought low. Strobing each of 128 row-addresses with $\overline{RAS}$ will cause all bits in each row to be refreshed. Further $\overline{RAS}$ -only refresh results in a substantial reduction in power dissipation.

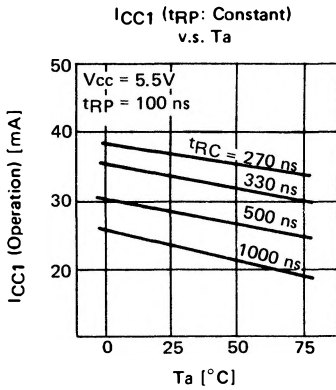
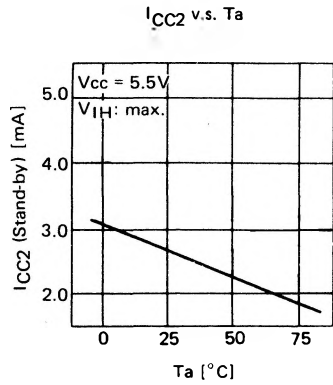
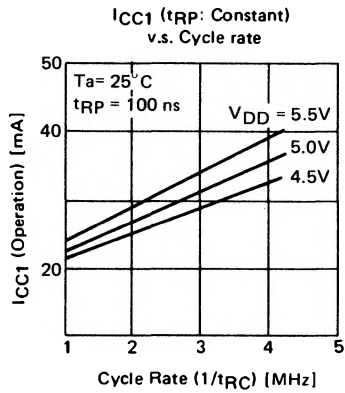
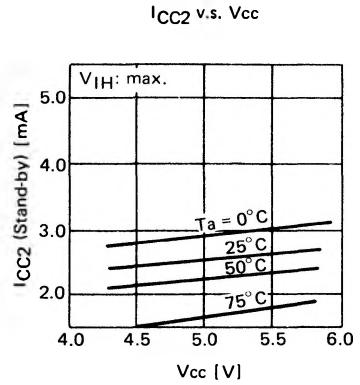
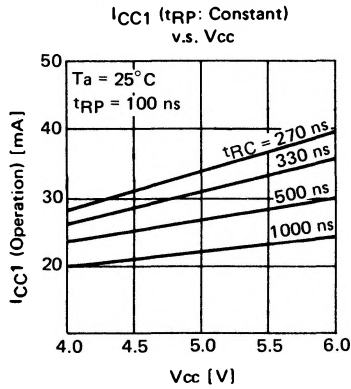
Hidden Refresh:

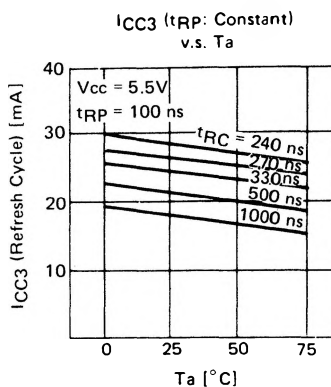
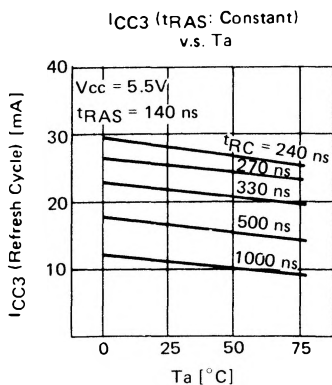
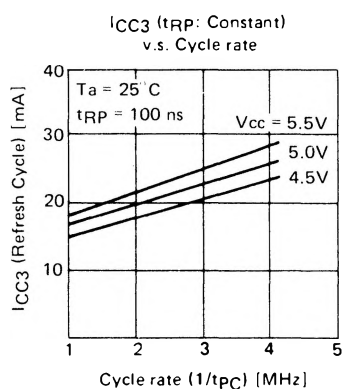
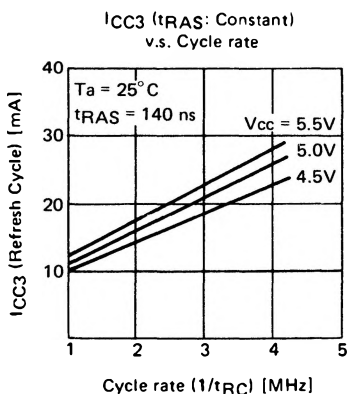
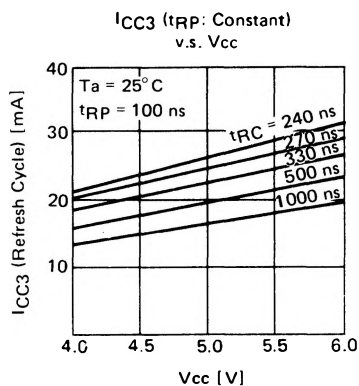
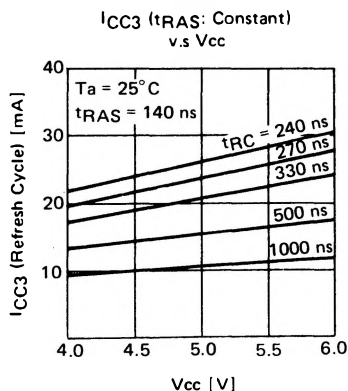
RAS ONLY REFRESH CYCLE may take place while maintaining valid output data. This feature is referred to as Hidden Refresh.

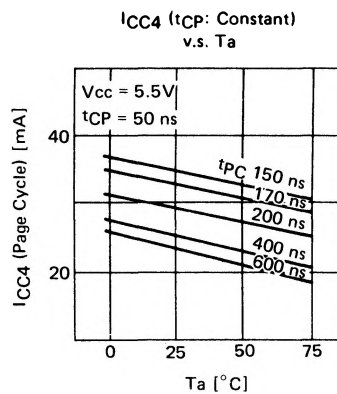
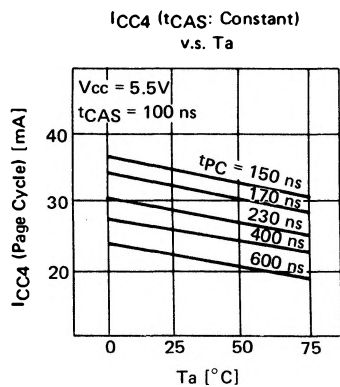
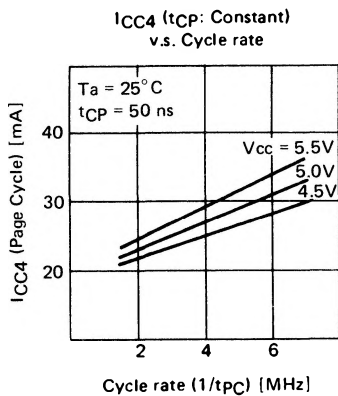
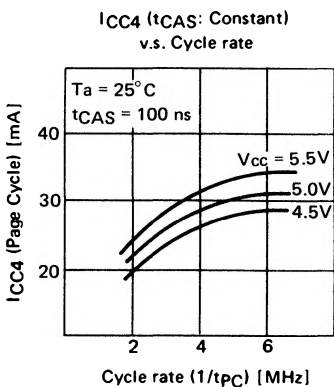
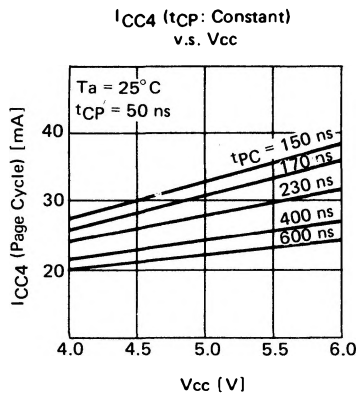
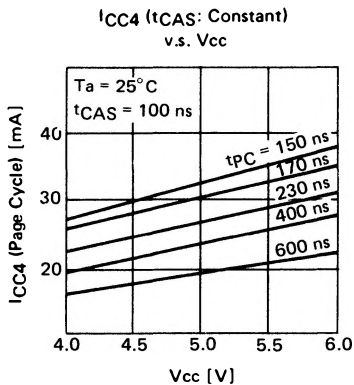
Hidden Refresh is performed by holding $\overline{CAS}$ as V_{IL} from a previous memory read cycle.

TYPICAL CHARACTERISTICS

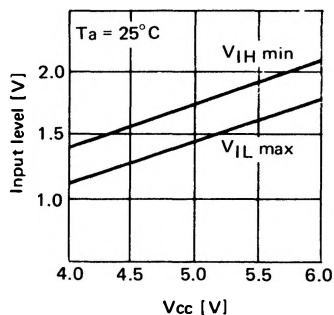




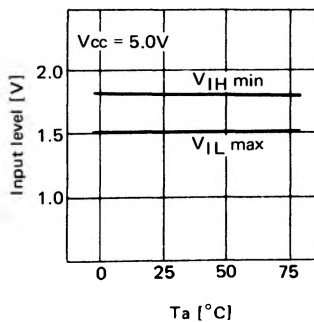




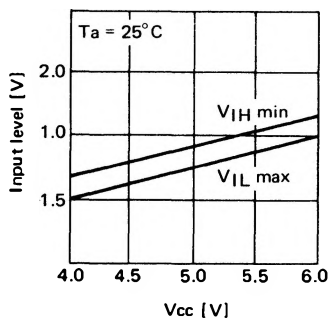
Address Input
v.s. V_{CC}



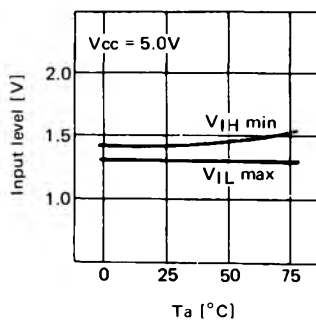
Address Input
v.s. T_a



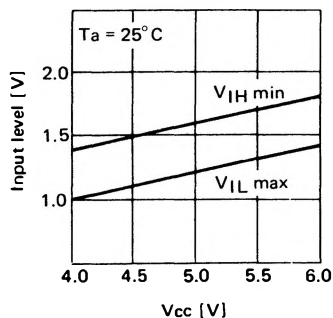
Data Input
v.s. V_{CC}



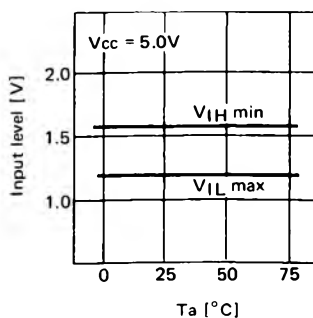
Data Input
v.s. T_a

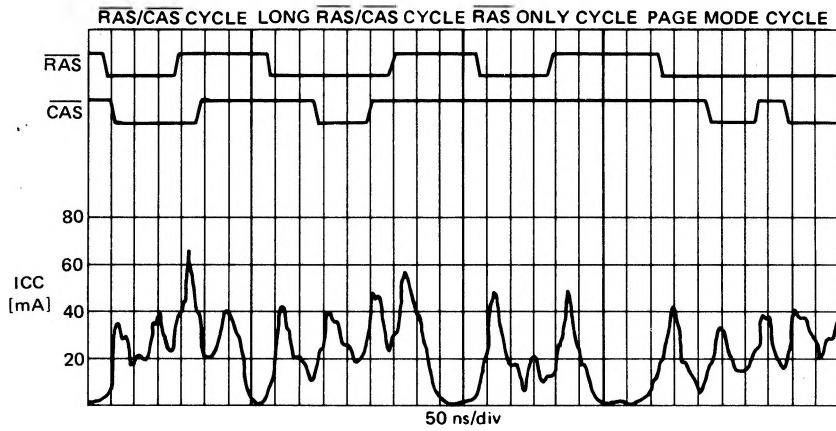


Clock Input
v.s. V_{CC}



Clock Input
v.s. T_a





MSM3732 Bit MAP (Physical-Decimal)

